Assignment No.4 : Socket Programming using C/C++/Java.

- **TCP Client, TCP Server**
- **UDP Client, UDP Server**

Code:

- **TCP Client, TCP Server:**

- **Server:**

```
import java.io.*;
import java.net.*;

public class Server {
    public static void main(String[] args) {
        int port = 12345; // Port to listen on

        try (ServerSocket serverSocket = new ServerSocket(port)) {
            System.out.println("Server is listening on port " + port);

            while (true) {
                Socket clientSocket = serverSocket.accept();
                System.out.println("Client connected: " + clientSocket.getInetAddress());

                // Create a new thread to handle each client
                ClientHandler clientHandler = new ClientHandler(clientSocket);
                new Thread(clientHandler).start();
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}

class ClientHandler implements Runnable {
    private Socket clientSocket;
```

```
    private PrintWriter out;
    private BufferedReader in;

    public ClientHandler(Socket socket) {
        this.clientSocket = socket;
    }

    @Override
    public void run() {
        try {
            out = new PrintWriter(clientSocket.getOutputStream(), true);
            in = new BufferedReader(new InputStreamReader(clientSocket.getInputStream()));
            String message;

            while ((message = in.readLine()) != null) {
                System.out.println("Client says: " + message);
                out.println("Server received: " + message);
            }

            clientSocket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

- **Client:**

```
import java.io.*;
import java.net.*;

public class Client {

    public static void main(String[] args) {

        String serverAddress = "127.0.0.1"; // Server IP address
        int serverPort = 12345; // Server port
        try (Socket socket = new Socket(serverAddress, serverPort);
            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
            BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
            BufferedReader reader = new BufferedReader(new InputStreamReader(System.in)) ) {
           System.out.println("Connected to the server.");

           while (true) {
               // Create a simple menu for the client
               System.out.println("Menu:");
               System.out.println("1. Send a message to the server");
               System.out.println("2. Quit");
               int choice = Integer.parseInt(reader.readLine());

               if (choice == 1) {
                   System.out.print("Enter a message: ");
                   String message = reader.readLine();
                   out.println(message);
                   String response = in.readLine();
                   System.out.println("Server response: " + response);

               } else if (choice == 2) {
                   System.out.println("Goodbye!");
                   break;
```

```
            } else {
                System.out.println("Invalid choice. Try again.");
            }
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
  }
}
```

Output:

Server:

```
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air OUTPUTS % javac Server.j
ava Client.java

(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air OUTPUTS % java Server
Server is listening on port 12345
Client connected: /127.0.0.1
Client says: Hello! TCP
Client says: Lets stop TCP
```

Client:

```
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air OUTPUTS % java Client
Connected to the server.
Menu:
1. Send a message to the server
2. Quit
1
Enter a message: Hello! TCP
Server response: Server received: Hello! TCP
Menu:
1. Send a message to the server
2. Quit
1
Enter a message: Lets stop TCP
Server response: Server received: Lets stop TCP
Menu:
1. Send a message to the server
2. Quit
2
Goodbye!
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air OUTPUTS %
```

Code:

- **UDP Client, UDP Server:**

- **Server:**

```
import java.net.DatagramPacket;
import java.net.DatagramSocket;

public class UDPServer {
    public static void main(String[] args) {
        try {
            // Create a DatagramSocket to listen on a specific port
            int port = 9876;
            DatagramSocket serverSocket = new DatagramSocket(port);

            System.out.println("UDP Server is running on port " + port);

            byte[] receiveData = new byte[1024];

            while (true) {
                // Create a DatagramPacket to receive data
                DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);

                // Receive data from the client
                serverSocket.receive(receivePacket);

                // Extract the received message and its source address and port
                String message = new String(receivePacket.getData(), 0, receivePacket.getLength());
                String clientAddress = receivePacket.getAddress().getHostAddress();
                int clientPort = receivePacket.getPort();

                System.out.println("Received from " + clientAddress + ":" + clientPort + ": " + message);

                // You can process the received message here

                // Respond to the client (optional)
                String response = "Server received: " + message;
                byte[] responseData = response.getBytes();
                DatagramPacket responsePacket = new DatagramPacket(responseData, responseData.length,
receivePacket.getAddress(), receivePacket.getPort());
                serverSocket.send(responsePacket);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

- **Client:**

```
import java.net.DatagramPacket;
import java.net.DatagramSocket;
import java.net.InetAddress;
public class UDPClient {
    public static void main(String[] args) {
        try {
            // Create a DatagramSocket for the client
            DatagramSocket clientSocket = new DatagramSocket();
            // Server address and port
            InetAddress serverAddress = InetAddress.getByName("localhost");
            int serverPort = 9876;
            String message = "Hello, UDP Server!";
            byte[] sendData = message.getBytes();
            // Create a DatagramPacket to send data to the server
            DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length, serverAddress,
serverPort);
            // Send the data to the server
            clientSocket.send(sendPacket);
            // Receive a response from the server
            byte[] receiveData = new byte[1024];
            DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);
            clientSocket.receive(receivePacket);
            String response = new String(receivePacket.getData(), 0, receivePacket.getLength());
            System.out.println("Received from server: " + response);
            // Close the socket
            clientSocket.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

Server:

```
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air 05B % java UDPClient
Received from server: Server received: Hello, UDP Server!
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air 05B %
```

Client:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air 05B % javac UDPServer.ja
va UDPClient.java
(base) utkarshkashinathtakmoge@Utkarshs-MacBook-Air 05B % java UDPServer
UDP Server is running on port 9876
Received from 127.0.0.1:49701: Hello, UDP Server!
```